

Exploring the solution space of the 1-Dollo phylogeny problem

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Introduction

When multiple, different SNVs occur within the same tumor, this is known as intra-tumor heterogeneity[3].

Phylogeny estimation algorithms that employ appropriate evolutionary models are key to understanding the evolutionary mechanisms behind intra-tumor heterogeneity[4].

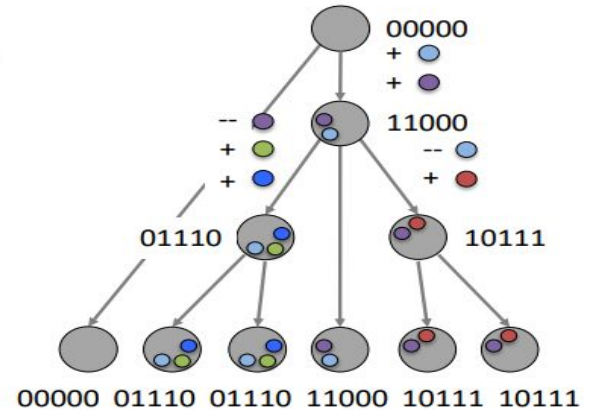
k-Dollo is such an evolutionary model, where a SNVs can only be gained once but lost k times.

Problem Statement

k -Dollo Phylogeny (k -DP) Problem

Definition 1. A k -Dollo phylogeny T is a rooted, node-labeled tree subject to the following conditions.

1. Each node v of T is labeled by a vector $\mathbf{b}_v \in \{0, 1\}^n$.
2. The root r of T is labeled by vector $\mathbf{b}_r = [0, \dots, 0]^T$.
3. For each character $c \in [n]$, there is exactly one *gain edge* (v, w) in T such that $b_{v,c} = 0$ and $b_{w,c} = 1$.
4. For each character $c \in [n]$, there are at most k *loss edges* (v, w) in T such that $b_{v,c} = 1$ and $b_{w,c} = 0$.

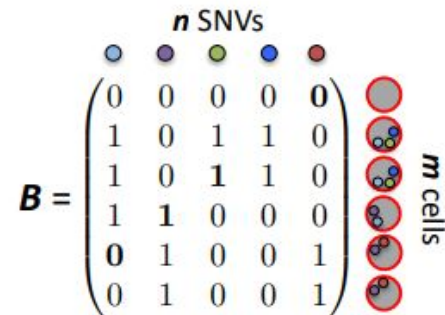


This picture is obtained from Dr. El-Kebir's slides:

https://www.el-kebir.net/teaching/CS598MEB//Spring_2020/lecture6.pdf

Introduction

k -Dollo Phylogeny problem (k -DP). Given a binary matrix $B \in \{0, 1\}^{m \times n}$ and parameter $k \in \mathbb{N}$, determine whether there exists a k -Dollo phylogeny for B , and if so construct one.


$$B = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

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Objective

The hardness of deciding whether B admits a k -Dollo completion A is open.

I am aiming to explore the solution space of 1-dollo completion.

Fundamental Theorem of Arithmetic

Every integer greater than one either is a prime number or can be uniquely represented by the product of primes.

For example : $6 = 2 \times 3$, $15 = 3 \times 5$, $210 = 2 \times 3 \times 5 \times 7$

-> This entails the prime encoding will give a unique integer based on presence or absence of prime

If we can represent SNVs with primes then we can have unique product for each cells.

Problem

	2	3	5	7		
Cell 1	1	0	1	0	$2^1 \times 3^0 \times 5^1 \times 7^0$	10
Cell 2	0	1	0	1	$2^0 \times 3^1 \times 5^0 \times 7^1$	21
Cell 3	1	1	0	0	$2^1 \times 3^1 \times 5^0 \times 7^0$	6
Cell 4	1	1	1	1	$2^1 \times 3^1 \times 5^1 \times 7^1$	210

GCD: Greatest Common Divisor

1. GCD of two numbers gives the product of all the common prime factors between them.
2. If there are no common prime factors then the $\text{GCD}=1$

$$\text{GCD}(10,21)=1$$

$$\text{GCD}(10,6)=2$$

GCD: Greatest Common Divisor

$$\text{GCD}(10, 21) = 1$$

$$\text{GCD}(10, 6) = 2$$

$$\text{GCD}(10, 210) = 10$$

GCD: Greatest Common Divisor

GCD between two cell_products will give the common SNVs between these cells.

Unique prime/ SNVs

1. In order to find the unique SNVs between cell_1 and cell_2

- $UNK_1 = \text{Cell_product_1} \setminus \text{GCD}(\text{cell_product_1}, \text{cell_product_2})$
- $UNK_2 = \text{Cell_product_2} \setminus \text{GCD}(\text{cell_product_1}, \text{cell_product_2})$

Divide the product of the cell with the GCD of two cells to find the unique prime/ SNVs of that cell with respect to another cell

Unique prime/ SNVs

B=

Primes →	2	3	5	7	Product
cell1	1	1	0	0	6
cell2	1	0	1	0	10
cell3	0	1	0	1	21
cell4	1	1	1	1	210

$$\begin{aligned} \text{gcd-1} &= \text{GCD}(6, 10) = 2 \\ \text{gcd-2} &= \text{GCD}(6, 21) = 3 \\ \text{gcd-3} &= \text{GCD}(6, 210) = 6 \\ \text{gcd-4} &= \text{GCD}(10, 21) = 1 \\ \text{gcd-5} &= \text{GCD}(10, 210) = 10 \\ \text{gcd-6} &= \text{GCD}(21, 210) = 21 \end{aligned}$$

UNK-X-Y = Unique prime mutation between cell-X and cell-Y

$$\begin{aligned} \text{UNK-1-2} &= 3 \\ \text{UNK-2-1} &= 5 \\ \text{UNK-1-3} &= 2 \\ \text{UNK-3-1} &= 7 \\ \text{UNK-1-4} &= 1 \\ \text{UNK-4-1} &= 35 \end{aligned}$$

$$\begin{aligned} \text{UNK-2-3} &= 10 \\ \text{UNK-3-2} &= 21 \\ \text{UNK-2-4} &= 1 \\ \text{UNK-4-2} &= 21 \\ \text{UNK-3-4} &= 1 \\ \text{UNK-4-3} &= 10 \end{aligned}$$

Translation

In order to change `cell_product_1=10` into `cell_2 _product_2= 21`

There are two actions required:

Gain: `Cell_product_1` needs to gain 21

Loss: `Cell_product_1` needs to lose 10

Translation

Loss: This is the Unique SNVs/prime of cell_product_1 with respect to cell_product_2 . UNK_1

Gain: This is the Unique SNVs/prime of cell_product_2 with respect to cell_product_1. UNK_2

This means Cell_1 has to lose its unique SNVs and gain unique SNVs of cell_2.

Reverse Translation

1. In order to change `cell_product_2 = 21` to `cell_product_1 = 10`
`Cell_product_2` has to lose `UNK_2` and gain `cell_product_1`.

This means Gain from `cell_1` to `cell_2` is the loss from `Cell_2` to `Cell_2` and vice versa.

Reverse Translation

1. In order to change `cell_product_2 = 21` to `cell_product_1 = 10`
`Cell_product_2` has to lose `UNK_2` and gain `cell_product_1`.

This means Gain from `cell_1` to `cell_2` is the loss from `Cell_2` to `Cell_2` and vice versa.

Underlying structures

1- dollo phylogeny problem can be formulated as:

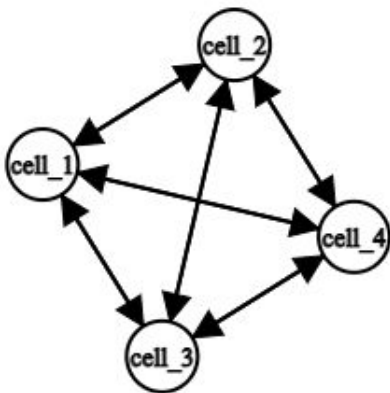
Can we cover all cell_products with limited loss and gains?

Limited loss: one division per prime.

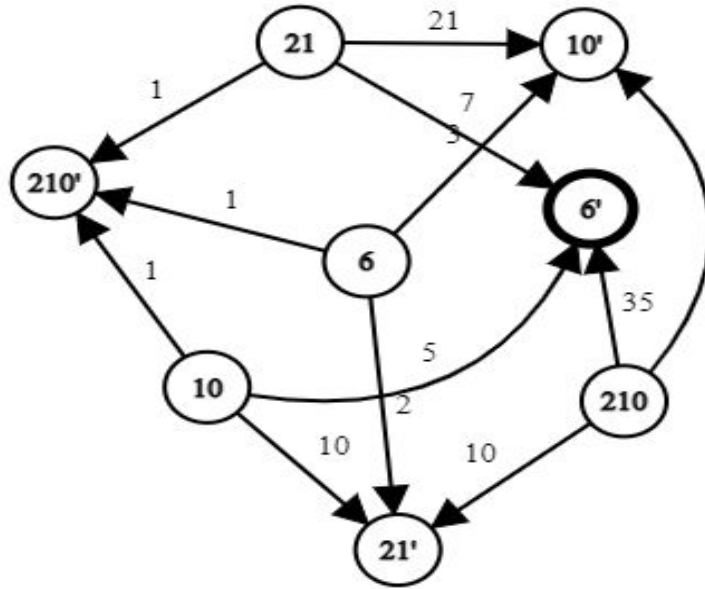
Limited Gain: one multiplication per prime.

Forming a Directed Graph

Directed Arrow Represents: Loss



Example



10', 21', 6' and 210' represents the same cell while coming from some other cell

10, 21, 6, 210 represents the cells which can form some other cell.

Example

i represents rows; j represents columns

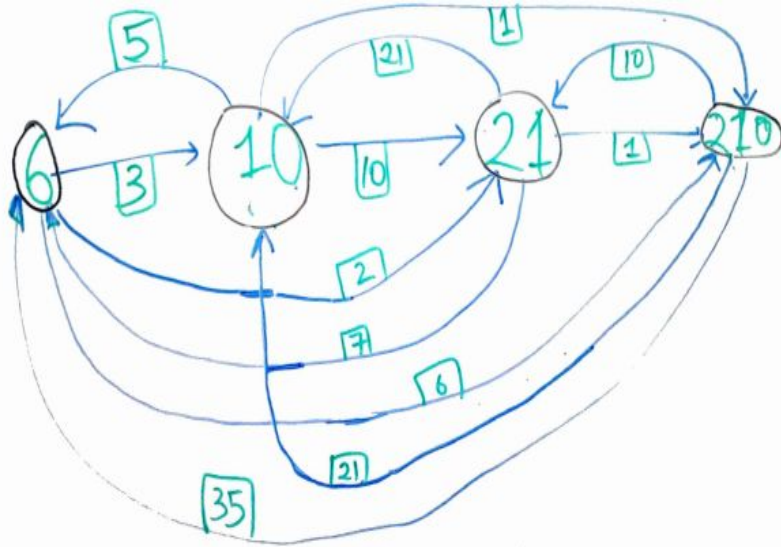
M =

X	6	10	21	210
6	1	3	2	1
10	5	1	10	1
21	7	21	1	1
210	35	21	10	1

$M[i][j]$ = Prime to loose from $[i \rightarrow j]$

$M[j][i]$ = Prime to gain from $[i \rightarrow j]$

Restating the problem



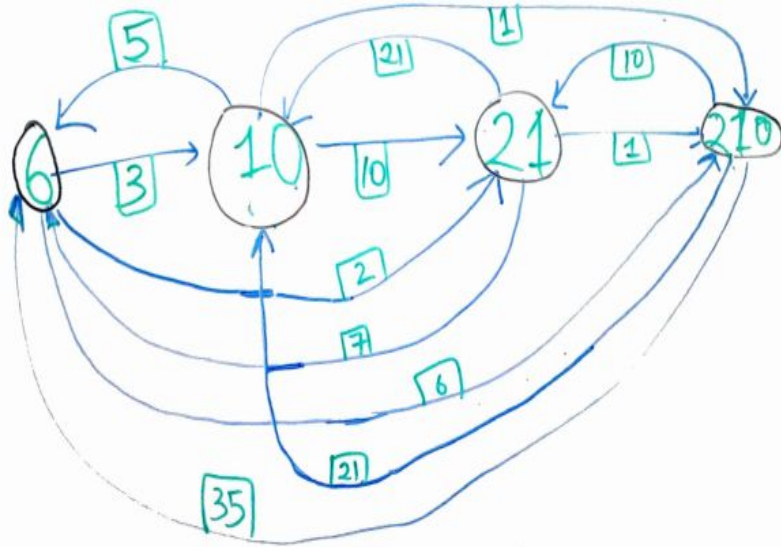
$G=(V,E)$

$V=\{6,10,21,210\}$

$E=\{(10,21),(21,10),(10,6),(6,10),(10,210),(210,10),(21,6),(6,21),(21,210),(210,21),(6,210),(210,6)\}$

Weights From Matrix M.

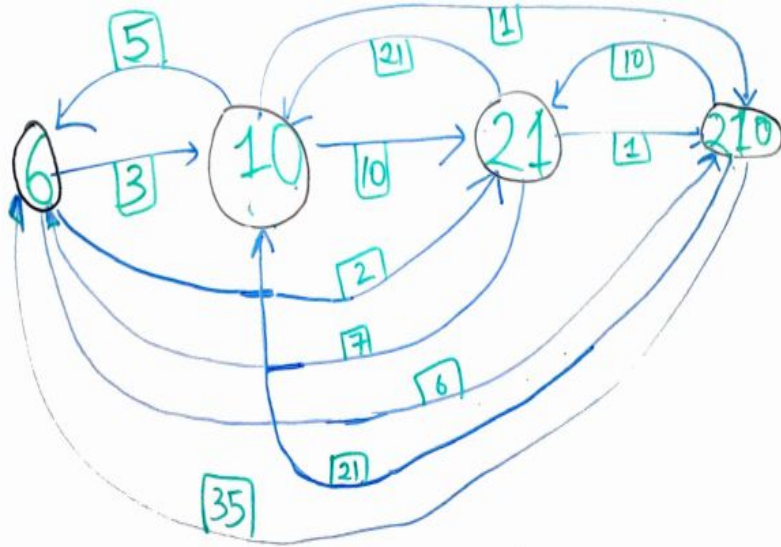
Restating the problem



If we can determine all the vertices are not reachable from vertex v belonging to V with sequence of losses and gains within the limited losses and gains constraints then the given matrix B does not have 1-dollo completion

This can be solved by DFS

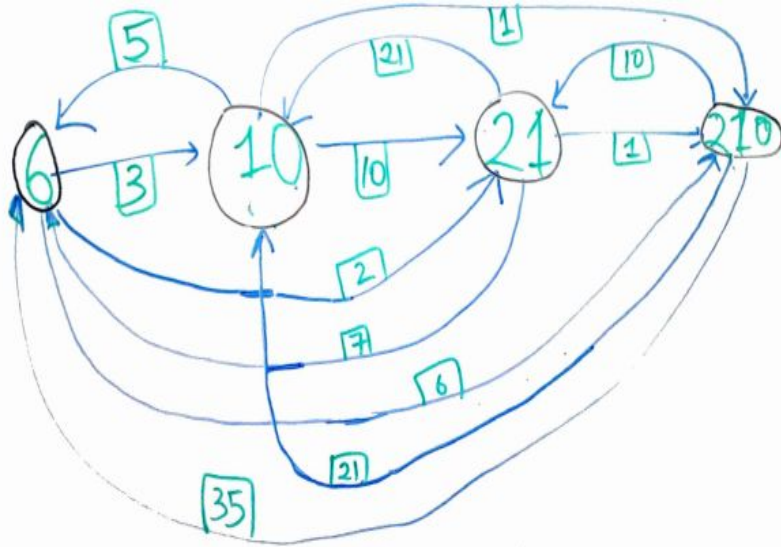
Restating the problem



Gain Constraint can be checked by checking if the gain does not belong to the primes/SNVs that was lost in the path.

Loss Constraint can be checked by checking if $M[i][j]$ divides $210 = 2 \times 3 \times 5 \times 7$ and with each loss we updated 210 by $210 / M[i][j]$

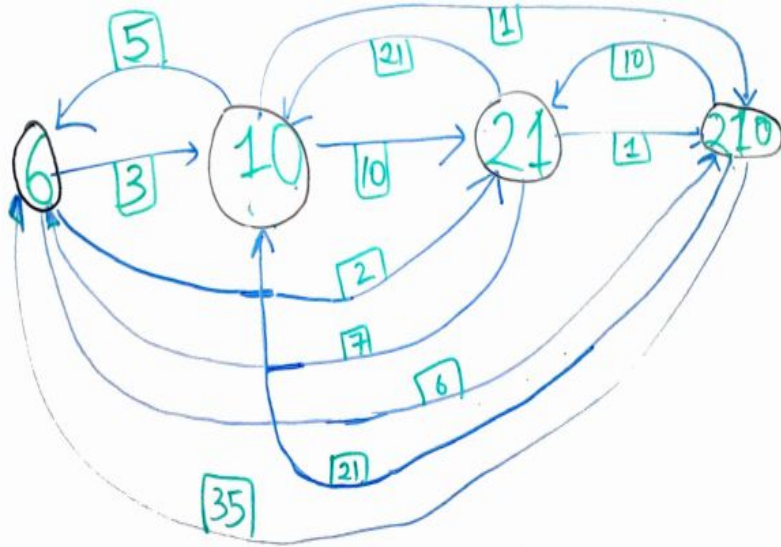
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Restating the problem

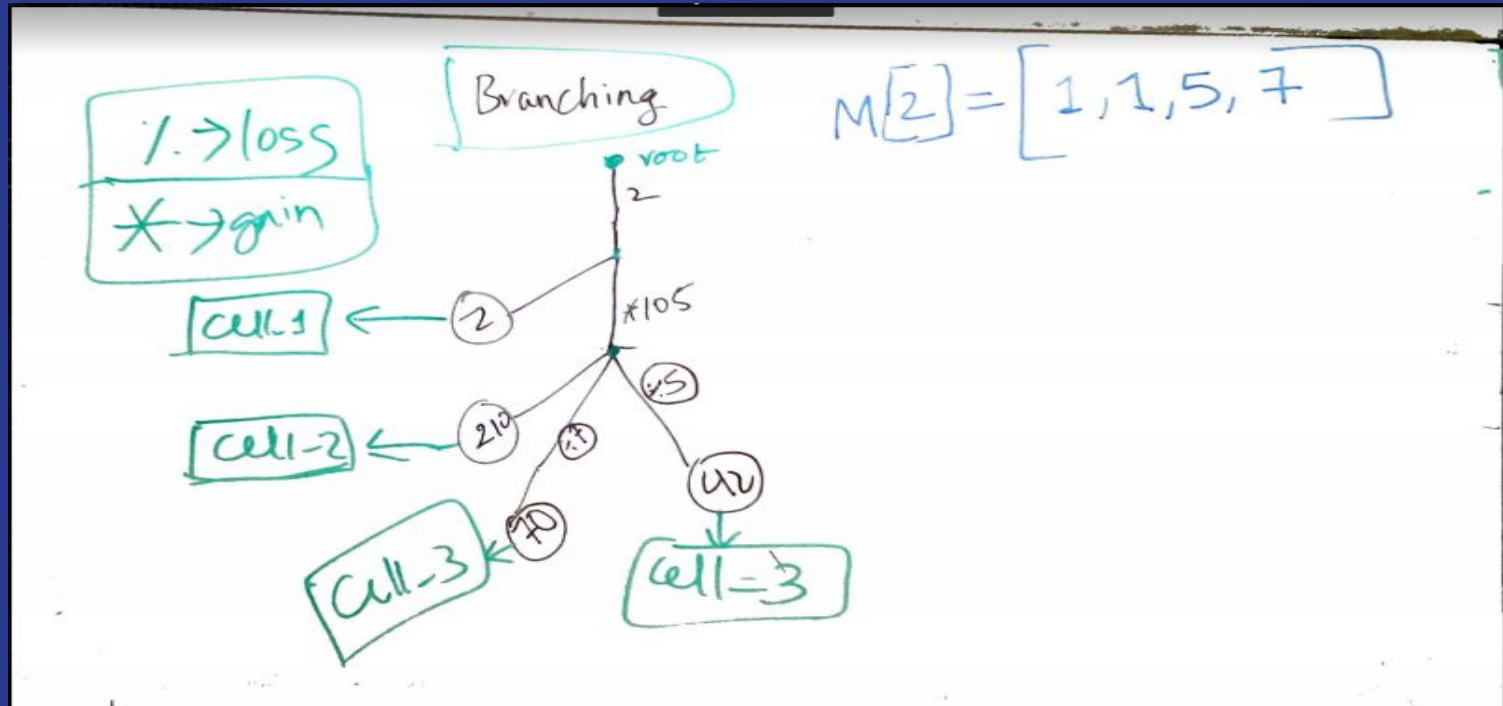


If all the nodes are not reachable, we can be sure that matrix B does not have 1-dollo completion.

But if it is reachable, then we cannot be sure if B has 1-dollo completion.

Since this is only checking for reachability we need to check for branching conditions.

Branching Condition



Branching Condition

1. If the starting vertex for DFS is 6 then for $M[6]$ we check if the losses are conflicting .
2. A conflicting loss is defined as follows .
 - A. If there are two losses in $M[6]$ that are co-primes to each other $\text{GCD}(\text{loss}_1, \text{loss}_2) = 1$ and
 - B. If there is a loss_3 in $M[6]$ such that both $\text{loss}_1, \text{loss}_2$ have $\text{GCD}(\text{loss}_1, \text{loss}_3) \neq 1$ and $\text{GCD}(\text{loss}_2, \text{loss}_3) \neq 1$

Results And Future Work

Results: Completed the work for 4×4 and 4×5 sizes.
Data generated by getting all 16 bit bin strings and partitioning it in groups of 4.

Future Work: Prove the conjecture and write a program to compute for any $m \times n$ and hopefully for any k .
And may be enumerate using polya enumeration.